

PEDRO BARBOSA

Algorithmic Support Engineer — Market Data, Python, SQL

Rio de Janeiro, Brazil (UTC-3, full overlap with US EST mornings) · Remote
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SUMMARY

Engineer who works on both sides of this role: I build and debug production systems, and I trade. I develop and backtest systematic strategies in Python on QuantConnect's LEAN engine, and I trade forex with my own capital through Interactive Brokers, so I understand intraday workflows, market data quality and execution realities — spread, slippage, gaps and missing ticks — from experience, not from a manual. On the technical side, my daily work is exactly first-line diagnosis: reproduce the reported problem, read logs and API responses, find the root cause, resolve it, and document cause and solution so the next person does not start from zero.

EXPERIENCE

Quantitative Developer — QuantConnect (freelance, remote)

September 2025 – present

- Build, backtest and run systematic multi-asset strategies in Python on the LEAN engine, over intraday and daily time-series data.
- Factor research, risk modelling and portfolio construction; live execution against broker APIs through custom scripts.
- Routinely investigate data anomalies: gaps, misaligned timestamps, corporate-action effects and survivorship issues that silently change results if nobody catches them.

Full Stack Developer — TIIX (remote)

April 2026 – present

- One of two developers on a healthcare platform used daily by patients, clinics and professionals.
- First-line and escalated incident work: reproduce the reported failure, investigate application logs, database state and REST API responses, identify the root cause, fix it, and add an automated check so it cannot return silently.
- Document every change with cause, solution and evidence — over a thousand written change records to date.
- Own features end to end: relational schema and versioned migrations, REST endpoints, interface and tests. Wrote 23 of the repository's 50 database migrations.
- Translate technical problems for non-technical stakeholders: support, product and business teams.

Independent Trader — Interactive Brokers

2024 – present

- Trade forex with personal capital; profitable operation with a written process for entries, risk limits and review.
- Day-to-day exposure to market data feeds, order execution, latency and the difference between backtest assumptions and live behaviour.

Python Developer — Freelance via Workana (remote)

January 2025 – March 2026

- Django systems and Python data pipelines for clients: access control, privacy-compliant data handling, normalised relational schemas.

Software Development Intern — TT&T Soluções em Informática (remote)

August 2024 – January 2025

- Django backend with payment gateway integration; MySQL modelling and documentation; unit tests with pytest; optimisation of slow queries and requests.

SELECTED WORK

QuantStrata — risk and exposure engine in Python, built so incomplete evidence can never be reported as zero risk: when the data does not cover the window, it refuses to produce a number instead of quietly assuming a flat day. 70 test cases, containerised pipeline, merged through formal code review.

ioide — open source daemon in Go: runs external adapters under a versioned contract with bounded timeouts, response validation, and a job queue with backoff and orphan recovery. 5,100 lines of code covered by 5,675 lines of tests.

SKILLS

Markets: systematic and algorithmic trading workflows, intraday and daily data, backtesting, risk metrics (drawdown, exposure, volatility), forex execution, market data quality analysis

Technical: Python (pandas, scikit-learn, pytest), SQL and relational databases (MySQL, PostgreSQL, SQLite), REST APIs, Linux, Docker, Git, log analysis and troubleshooting

Support practice: incident reproduction, root-cause analysis, written documentation of causes and resolutions, clear communication with technical and non-technical users

Cloud: AWS EC2 — provisioning and operating Linux instances, deploying services and troubleshooting them in a cloud environment

EDUCATION AND CERTIFICATIONS

Computer Engineering (Data and Machine Learning) — Centro Universitário IBMR · 2023 – 2028 (expected) · GPA 8.7/10

Scientific initiation in Economic Mathematics and Machine Learning certificate — IMPA, 2024

HackerRank: Problem Solving (Intermediate), REST API (Intermediate), Python · Coursera: Python for Data Science and AI